NSF Research Proposal:

Background: Over the past two decades, precipitous advancements have characterized the field of artificial intelligence, specifically in the field of deep learning, where myriad new architectures of artificial neural networks (ANNs) have been developed. While traditional networks take the form of a discrete-layered structure, recent innovations have spawned interest in continuous, ordinary differential equation (ODE)-controlled versions of residual networks (ResNets) and graph neural networks (GNNs) [1].

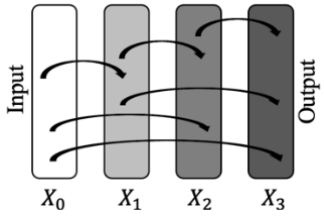


Figure 1: DenseNet connects each discrete layer to every layer that follows it.

Continuous networks have shown many advantages over discrete models, many of which relate to the flexibility with which one can alter the model before, during, and after training [2]. One highly interesting result of continuous ANNs is the **decoupling of depth and the number of parameters**. While in a discrete network these two quantities are invariably tied together, in continuous models it is possible to wholly change one without the other [2], a process generally known as *manifestation invariance* [2].

$$\boldsymbol{X}(t) = \boldsymbol{F}\left(\alpha(t)\boldsymbol{F}^{-1}(\widehat{\boldsymbol{X}_0}) + \int_0^t \boldsymbol{\Theta}(s,t)\boldsymbol{X}(s)\mathrm{d}s + \boldsymbol{B}(t)\right)$$

Figure 2: IE for computing a layer X at any continuous depth t. F is an invertible activation function (such as leaky ReLU), $\Theta(s,t)$ is a matrix of weights from layer s to t, B are biases, and α is the injection scaling function.

One discrete architecture which has not been converted into a continuous system is DenseNet [3]. DenseNet is a convolutional network that densely connects each layer to all following layers. This discrete architecture has proven very effective at image classification, requiring far fewer parameters to achieve an even lower error rate than state-of-the-art image classifiers [3]. While most ResNets follow a natural conversion into a system of ODEs, DenseNet's continuization naturally produces a system of integral equations (IEs). It is not clear what advantages and disadvantages may come in an IE-controlled ANN when compared to those controlled with ODEs. **I will implement and test a novel continuous, IE-controlled version of DenseNet and investigate the parameter-depth decoupling.**

I am markedly qualified to act as a researcher in this project for the following reasons: (1) **I have already begun this project with the leading researchers** (S. Scott, M. Foss, P. Radu) to convert a toy model into a GPU-enabled Python class that generalizes all parameters of the model. I have also tested this model's regression strength on small datasets. (2) Having done previous research on models that perform animal classification (M.R. Hasan), I have considerable knowledge in methods for evaluating classifiers and in improving accuracy. (3) Understanding the mathematics behind the model, I will be able to conduct analytic tests (and not just empirical tests) that confirm its convergence based on constructed inputs.

Previous Work: As can be seen in Figure 1, DenseNet works by connecting each layer to all subsequent layers. Note that in the discrete model, $X_0, X_1, \ldots, X_n$ is each a vector of values. If we let $X(t)$ be a vector-valued function that represents the activations at depth t, we can take the limit as $\Delta t \to 0$ and get a continuous function $X(t)$. For any given depth t, we sum over the product of all weights and states of previous layers, giving us an integral, shown in Figure 2.

An issue we discovered was that if $X(t)$ is formulated only as the integral term, then the solution does not depend on the input. To solve this, we introduce the "injection" term, $\alpha(t)F^{-1}(\widehat{X_0})$which injects some factor of the input $\widehat{X_0}$ at every step according to scaling function α. For example, α can be a function that goes to 0 so as to reduce dependence on the input at deeper layers. This "injection" can be seen in Figure 3.

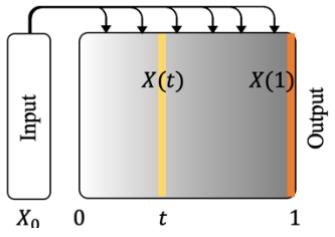


Figure 3: IE DenseNet computes the integral of all previous layers at each step and injects the input. The output is

Each element $\Theta_{i,j}(s,t)$ can be written as a linear combination of some set of K nonlinear basis functions, weighted by tunable parameters α_{ij}^{ℓ} and β_{ij}^{ℓ}, $1 \le \ell \le K$. K directly increases the **number of parameters**, and in principle could increase the complexity of problem that the model can solve.

During feedforward, we use any numerical integrator, where the number of iterations determines the **depth**. The depth affects the quality of the numeric solution to Figure 2, and may need to increase with K, despite being decoupled (as with ODEs). Current literature for IE ANNs does not explain the relationship between these two factors.

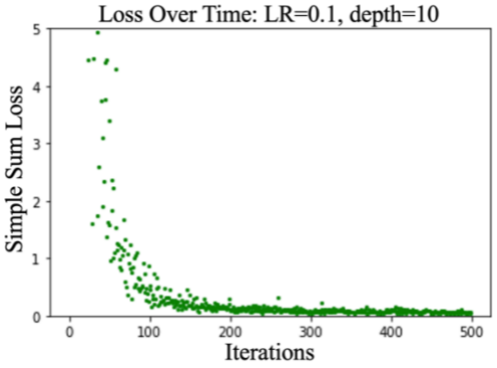


Figure 4: Noise-correction test shows that loss rapidly decreases with a learning rate of 0.1 and depth of 10.

Plan of Work: **(1)** Implement the architecture in Python using TensorFlow. **I have successfully accomplished this.** To compare our model to state-of-the-art models, we need to ensure that feedforward and backpropagation are GPU-enabled. To this aim, I rewrote the model's operations using Tensor-Flow's tensors and a gradient tape, which allows for speedy training on GPUs. **(2)** Convert the model from a regression model into a classifier. The current model is capable only of outputting a tensor of the same shape as the input. To correct this, I will add connected layers and an output layer. **(3)** Compare the trained model to state-of-the-art models on benchmark datasets (MNIST, CIFAR-10, CIFAR-100, SVHN, ImageNet). Preliminary tests that I have conducted on small regression problems show that the model is capable of quickly reducing loss. One test that I performed was noise correction, in which the model was fed a random matrix of noisily perturbed 1's and 0's and attempted to remove the noise. Figure 4 shows how the model quickly minimized loss. However, these preliminary tests likely did not measure the full strength of the model since each input was independent of surrounding data (unlike image data), and thus could not take advantage of the model's convolution. I will test the model on benchmark image datasets and compare it to models such as ResNet-200, ODE-Net, DCN-ODE, and the original, discrete DenseNet [4][5]. I will then perform a full analysis of accuracy using set parameters. **(4)** Investigate the relationship between the model's problem-solving capabilities and the parameter-depth decoupling. While some analysis of this decoupling has been done in ODE networks [2], it is unclear how it will translate to an IE network. Specifically, I will observe how these hyperparameters affect decision boundaries, and I will experiment with techniques of 'bottlenecking' the network and continuously reducing its width as a function of depth.

Intellectual Merit: Since continuous networks are an intense area of research in the field of machine learning, development of new IE models is key in filling knowledge gaps about the strengths and advantages of continuized ANNs. Likewise, the parameter-depth decoupling is poorly understood—particularly for IE-based models, leaving much room for discovery. Finally, the techniques I will develop to perform analyses and optimization of these ANNs will be highly relevant to any research of continuized networks [4].

Broader Impacts: Continued advancements in ANNs have the potential to influence nearly every facet of human technology. As many machines become smaller, the need for loading working ANN models that can be trained and used with minimal computation is increasingly imperative. Having the flexibility of continuized networks allows for engineers to tailor-make a model for any given problem. Because continuized networks can train with a large number of parameters and a high depth and then be 'compressed' into some desired number of parameters [2], they show much promise in everyday, real-time applications. For example, many autonomous machines would benefit from pre-trained models that solve image classification problems and maximize their performance given a set memory and decision time. **My work will directly lead to a better understanding of continuized networks** by testing a novel architecture and will lay the groundwork for future IE-based networks.

References: [1] Michael Poli et al. "Graph neural ordinary differential equations". In: arXiv:1911.07532 (2019). [2] Alejandro F Queiruga et al. "Continuous-in-depth neural networks". In: arXiv:2008.02389 (2020). [3] Gao Huang et al. "Densely connected convolutional networks". In: CVPR. 2017, pp. 4700–4708. [4] Ricky TQ Chen et al. "Neural ordinary differential equations". In: NeurIPS. Vol. 31. 2018. [5] Nergis Tomen et al. "Deep continuous networks". In: ICML. PMLR. 2021, pp. 10324–10335.